PLUMBING (Continued)

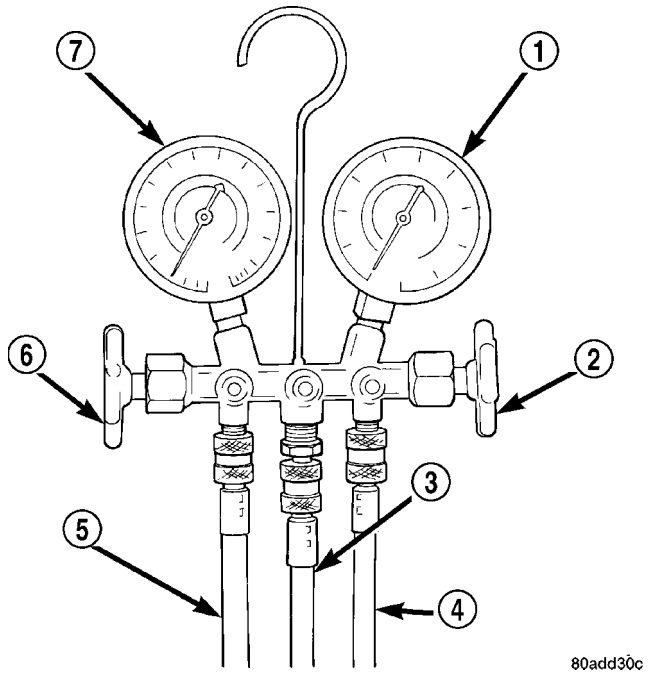

Fig. 2 Manifold Gauge Set - Typical

1 - HIGH PRESSURE GAUGE
2 - VALVE
3 - VACUUM/REFRIGERANT HOSE (YELLOW W/BLACK STRIPE)
4 - HIGH PRESSURE HOSE (RED W/BLACK STRIPE)
5 - LOW PRESSURE HOSE (BLUE W/BLACK STRIPE)
6 - VALVE
7 - LOW PRESSURE GAUGE

REFRIGERANT SYSTEM CHARGE LEVEL

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

NOTE: Always refer to the A/C Underhood Specification Label for the refrigerant fill capacity of the vehicle being serviced.

The following procedure should be used to determine whether the A/C refrigerant system contains the proper refrigerant charge. Symptoms of an improper refrigerant charge (low) include: poor A/C performance, fog emitted from the air outlets, a hissing sound from the A/C orifice tube/liquid line area. There are two different methods with which the refrigerant charge level may be tested:

1. Using a DRBIII® scan tool, a thermocouple and the Charge Determination Chart (Fig. 3). Refer to the appropriate diagnostic information.

2. Using a manifold gauge set, a thermocouple and the Charge Determination Chart (Fig. 3).

A temperature probe is required to measure liquid line temperature. The clamp-on, Type K thermocouple temperature probe used in this procedure is available through the DaimlerChrysler Professional Service Equipment (PSE) program. This probe (PSE #66-324-0014 or #80PK-1A) is compatible with temperature-measuring instruments that accept Type K thermocouples, and have a miniature connector input. Other temperature probes are available through aftermarket sources; however, all references in this procedure will reflect the use of the probe made available through the PSE program.

In order to use the temperature probe, a digital thermometer will also be required. If a digital thermometer is not available, an adapter is available through the PSE program that will convert any standard digital multimeter into a digital thermometer. This adapter is designed to accept any standard Type K thermocouple. If a digital multimeter is not available, this tool is also available through the PSE program.

NOTE: When connecting the service equipment couplings to the refrigerant system service ports, be certain that the valve of each coupling is fully closed. This will reduce the amount of effort required to make the connection.

(1) Remove the caps from the refrigerant system service ports and attach a manifold gauge set or a R-134a refrigerant recovery/recycling/charging station that meets SAE Standard J2210 to the refrigerant system.

(2) Attach a clamp-on thermocouple to the A/C liquid line. The thermocouple must be placed as close as possible to the inlet end of the A/C orifice tube to accurately observe liquid line temperature.

(3) Bring the refrigerant system up to operating temperature and pressure. This is done by allowing the engine to run at idle under the following conditions for five minutes.

(a) Front windows are open.

(b) Transmission in Park or Neutral with the parking brake set.

(c) A/C-heater controls set to fresh air, full cool, panel outlet mode, high blower speed and the A/C compressor engaged.

(4) Raise the liquid line (discharge) pressure to about 1793 kPa (260 psi) by placing a piece of cardboard over the front of the A/C condenser. To place the cardboard properly, remove the upper radiator sight shield (if equipped) and cover only enough of the A/C condenser to raise and maintain the liquid line pressure at the previously specified level.